

Can Eye-Tracking Capture Human Reasoning During Text Classification?

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Abstract

This empirical user study examined the efficacy of eye-tracking as a medium to capture human reasoning when they perform text classification. Each participant had to classify twelve injury narratives of varying complexity into six cause-of-injury categories. The participants' eye-tracking data including word-level fixation count and fixation duration were recorded while they performed text classification. The participants also verbally explained their reasonings for classification choices for each narrative and this think-aloud data was recorded and transcribed. To determine the efficacy of eye-tracking, we analyzed the overlap between most predictive narrative words identified using highest fixation count and fixation duration, with the narrative words mentioned in think-aloud reasoning. Overall, eye-tracking seemed to be a reliable way of capturing human reasoning during text classification as relatively consistent overlap between eye-tracking and think-aloud words was observed with varying conditions of narrative complexity, classification categories, classification time taken, and prediction accuracy.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; **Empirical studies in HCI**; **HCI design and evaluation methods**;

Keywords

Eye-tracking, Text Classification, Human reasoning, Language processing

ACM Reference Format:

JeevithaShree Divya Venkatesh, Aparajita Jaiswal, and Gaurav Nanda. 2026. Can Eye-Tracking Capture Human Reasoning During Text Classification?. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798352>

1 Introduction

Text classification has a wide range of applications in different domains including healthcare, social sciences, business intelligence, and content management in general. Most of these applications involve machine learning (ML) based automated approaches for text classification, which require training ML models on manually

classified/labelled textual data. Therefore, the accuracy and consistency of human labelled data are critical for these automated text classification applications to be effective. However, in real-world, variations and inconsistencies are commonly observed in human labeling; most likely due to difference in subjective interpretations of text by different individuals [1], [2]. For reliability and trustworthiness of these ML-based text classification approaches, explainability of ML models' output is important, and, for this purpose, different explainable AI approaches have been developed for ML models [3], [4]. However, the aspect of explainability has often been overlooked for the human classification/labeling tasks. In most human text classification tasks, only the final outcome in form of labelled data is collected, but the reasoning behind labelers' classification choices is not recorded as it can be time and resource consuming. However, the reasoning related data can be fairly valuable as it can be used to identify inconsistencies and also be embedded into subsequent AI-based practical systems, particularly in domains where the accuracy and reliability of predictions are critical, such as healthcare.

This study aims to fill this gap by examining the efficacy of eye-tracking as a method to understand human reasoning when people perform text classification, as eye tracking has been found to be a reliable method for studying human text comprehension [5], [6], [7], [8]. To this end, a text classification user study was conducted where participants had to read accident narratives and select the most applicable cause-of-injury code from a list of six unique codes. To validate the user reasoning behind text classification decisions, a think-aloud protocol was employed, where participants verbalized their thought processes and reasoning behind selecting a particular cause-of-injury code. The think-aloud approach helped in understanding how specific words in the narrative text influenced participants' classification decisions.

1.1 Significance of Text Classification in Injury Surveillance

The need for automated text classification in healthcare, specifically, injury surveillance is driven by the growing volume of unstructured textual data generated such as clinical notes, injury reports, and patient records. Efficient and accurate categorization of injury-related text would be helpful for improving outcomes in public health, clinical practice, and safety policy-making [1], [9]. Automated injury classification enables timely analysis of large volumes of raw unstructured textual data collected from hospitals facilitating the identification of injury patterns, risk factors, and trends in a consistent, efficient and scalable manner [2], [10], [11].



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ACM ISBN 979-8-4007-2281-3/2026/04

<https://doi.org/10.1145/3772363.3798352>

1.2 Related Work on Eye Tracking for Natural Language Processing

Eye tracking records eye movements and gaze patterns [12] and can offer insights into how individuals interact with and interpret textual information. By capturing data such as fixation duration, saccadic movements, and gaze paths, eye tracking provides detailed information about their reading process [7]. Eye-tracking has been used to analyze human text comprehension abilities as eye movements while reading is one of the most important psychological signals [13], [14]. Eye fixation data is a crucial ocular metric as it enables the human mind to actively engage with content and reveals what parts of the content the user is focusing on [8]. The analysis of the fixation patterns allows researchers to understand how individuals process information and the challenges they may face when engaging with complex content [13]. Additionally, the eye-tracker also provides information about other ocular parameters such as changes in pupil diameter or pupil dilation, which has been used as an indirect measure of cognitive load and emotional arousal experienced by the participants [5], [15]. Multiple studies have used eye-tracking data to study cognitive processing of the users for language processing tasks. For example, Yun (2021) [15] used eye-tracking to examine the comprehension of scientific terminology and its impact on cognitive processes during science reading among middle school students using ocular parameters such as first fixation duration, dwell time, and revisits. Similarly, other studies have used eye-tracking data for developing cognitive models for natural language processing (NLP) tasks such as reading and manual sentiment annotation complexity [5], [6], [12], [13]. Some of the recent studies [16], [17] have also used neural networks to integrate text features with eye-tracking data for improving various NLP tasks, such as determining sentiment polarity and detecting sarcasm. Beyond text comprehension, existing studies [18] have shown that eye movements serve as real-time indicators of reasoning, with gaze patterns reflecting how people process and integrate information during logical tasks. In visual and logical reasoning tasks [19], fixation durations, saccades, and pupil dilation systematically differentiate the high and low performing reasoners and signal key processes like conflict detection and problem-solving strategies [20], demonstrating that visual attention directly reflects cognitive reasoning [21].

2 METHODS

2.1 Experimental Design

The eye tracking user study was conducted at Purdue University after IRB approval with 51 participants in which they had to perform a text classification task of assigning six possible cause-of-injury codes to accident narratives. The dataset used for this study contained injury narratives collected at hospitals and associated cause-of-injury codes assigned by professional injury coders [22]. A computer interface was developed for the text classification task which displayed the injury narrative and presented the six cause-of-injury categories as buttons. Tobii Pro Fusion Eye Tracker was placed along the lower edge of the computer monitor, which displayed the injury classification software. Each participant had to read 12 accident narrative prompts (one at a time) on a computer

screen and select the most applicable cause-of-injury category for each prompt from six possible cause-of-injury codes: CUT, FALL, STRUCK, BURN, MOTORVEHICLE, and OTHER. Participants' ocular parameters were recorded by the eye-tracker while they performed text classification, including word-level fixation counts and fixation duration for each word in the narrative. Participants were also asked to explain their reasoning for classification verbally for each prompt. These think-aloud responses were recorded using an EVIDA digital voice recorder.

2.2 Data Collection and Analysis

For the dataset used in the user study, there were 17 unique prompt sets, with each prompt set containing 12 narrative prompts. In each prompt set, the 12 prompts were composed of 2 unique narratives belonging to each of the 6 cause-of-injury codes used in this study. For each prompt set, human text classification data was collected from 3 participants, totaling to 51 participants ($=17 \times 3$). SET1, SET2 and SET3 are referred to as the eye tracking data collected for 3 batches of cases (for each prompt set) across 51 participants. Thus, the overall data included: (a) total of 34 ($=17 \times 2$) unique narrative cases for each of the 6 cause-of-injury codes, (b) total of 204 ($=34 \times 6$) unique narrative cases (ranging from 3 to 62 words in length), and (c) total of 612 ($=204 \times 3$) data points collected from the eye-tracking study. The data points collected for each prompt for every participant included: (a) the cause-of-injury code selected by participant, (b) the total time taken by participant to perform the classification, (c) the eye-tracking recording from Tobii eye-tracker, and (d) the think-aloud audio recording. The eye tracking data recorded during participants' text classification processes was analyzed using Tobii Pro Lab software. Each individual word in the narrative was marked as a separate area-of-interest (AOI) and the fixation count and fixation duration were calculated for each AOI, i.e., each word in the narrative. The intuition behind this approach was that the words to which a participant paid most attention are likely to have the greatest influence in user's classification decision making process. Thus, the top words that influenced human text classification were selected based on highest fixation count and fixation duration values.

The think-aloud data collected during the study contained verbal explanations of participants' reasoning behind their classification choices. These explanations typically mentioned specific parts of the narrative (words and phrases) based on which the participant selected a particular category for the given narrative. The think-aloud audio data was transcribed and analyzed along with the eye tracking data to evaluate the alignment of eye-tracking data recorded during participant's text classification task. To examine the effectiveness of eye tracking in capturing human reasoning, we compared the overlap between the most important words identified by eye tracking (ET) based on the highest fixation count (FC) and fixation duration (FD) values, and the words from the injury narrative mentioned in think-aloud (TL) transcript. The level of overlap of words between the think-aloud transcript and eye tracking words (referred to as ET-TL overlap, with 99% data points ranging between 1-10 words) was qualitatively examined by researchers and each record was assigned one of the following three levels of overlap—*Low*, *Medium*, and *High*. The criteria used was: (a) if the ET-TL overlap was < 3

words, it was marked as *Low*, (b) if the ET-TL overlap was between 3-5 words, it was marked as *Medium*, and (c) if the ET-TL overlap was > 5 words, then it was marked as *High*.

The narratives were also pre-categorized by one of the researchers with expertise in injury classification into three complexity levels- *High*, *Medium*, and *Low*. The complexity level of each narrative was determined based on the difficulty of text comprehension and whether the cause of injury was obvious or not (largely due to multiple injuries in narrative or not enough information available in the narrative). Participants were not aware of these different narrative complexity levels during their task. Out of the total 204 unique injury narratives in the dataset, 74 narratives were categorized as “*Low*”, 82 narratives were categorized as “*Medium*”, and 48 narratives were categorized as “*High*” complexity. These results are presented in the next section.

3 RESULTS

In this section, we present the results from user study and discuss the insights obtained from data analysis. The time taken by participants to classify one narrative prompt was observed to be in the range of 10-90 seconds. For subsequent analysis, the classification time per narrative was grouped into three equally-divided time-range levels: a) *Low* (< 30 seconds), b) *Medium* (30-60 seconds), and c) *High* (> 60 seconds). The accuracy of participant’s prediction of the cause-of-injury category was evaluated based on the original cause-of-injury code in the dataset. If the cause-of-injury code selected by participant agreed with the original code, it was counted as an accurate/correct prediction. An example narrative case, associated data collected from the user study, and analysis results are shown below:

Narrative: *Painful lower back post his brother pushing off his back in the pool 24 hours ago panadeine forte given at 0800hrs with nil relief.* **Original Category in dataset:** OTHER

Complexity: High (as the cause of injury is not mentioned clearly in the narrative)

Predicted Category from 3 participants: STRUCK, FALL, OTHER

Data for the participant who predicted FALL:

Time Range: Medium (total time taken for classification: 42 seconds)

Think-aloud transcript: *The most appropriate answer here is fall because the pain is due to his brother pushing off him towards the pool. So it comes in the category of fall.*

Most important words identified from eye-tracking data based on highest fixation count and fixation duration: [*his, back, the, pushing, off, in, pool, hours, brother, 24, panadeine*]

ET-TL overlap level: High (6 overlapping ET and TL words: *his, brother, pushing, off, the, pool*).

We analyzed the variation in classification time with respect to narrative complexity. Figure 1 presents the distribution in percentage of cases of the three levels of classification time, for the three levels of complexity.

As shown in Figure 1, the low complexity narrative group had a relatively higher percentage of low classification time cases as compared to medium and high complexity groups. The medium complexity group had a relatively higher percentage of medium

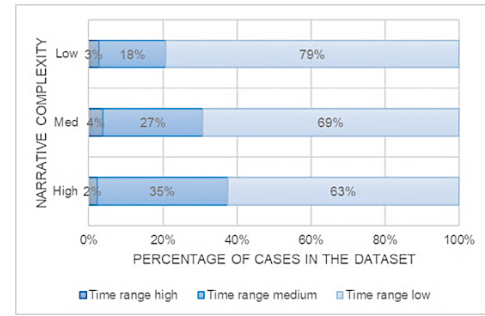


Figure 1: Relationship between Complexity of Narrative and Time taken for Classification

time range cases as compared to lower complexity group. Similarly, the high complexity group had an even higher percentage of medium time range cases as compared to medium complexity group. It is interesting to note that the percentage of high time range cases was not considerably different in low, medium, and high complexity groups, and relatively larger variation was observed between the distribution of low and medium time range cases.

Next, we studied the variation in the percentage of cases of high, medium, and low ET-TL overlap cases based on (a) the total time taken by participant to complete the classification task, (b) the accuracy of cause-of-injury codes assigned by participants, (c) complexity level of narratives, and (d) the different cause-of-injury codes/categories. These are presented in Figure 2 (a), (b), (c), and (d) respectively. Figure 2(a) depicts the distribution of high, medium, and low ET-TL overlap level cases for different groups of total classification time; the ET-TL overlap level had some variation based on the total time taken for classification. For all time range groups, the percentage of high ET-TL overlap cases was the highest, with relatively higher percentage (56%) observed for high time range cases. This might have been because the participants spent more time focusing on the important narrative words. However, it is to be noted that even for the low time range group, the percentage of high ET-TL overlap cases was reasonably similar (52%) indicating that eye tracking was able to reliably capture the words that participants focused on even for shorter duration. We also noticed a greater variation between the percentage of medium and low ET-TL overlap cases for medium and high time range groups. The high time range group had a relatively higher percentage of low ET-TL overlap cases (39%). One of the possible reasons for this might be that those narratives were complicated and even though the participants focused on certain narrative words for longer times, they concluded differently in their think-aloud reasoning. Another possibility is that with increased time spent on the narrative, the chances of having noise in eye-tracking data would also increase.

We then compared the distribution of high, medium, and low ET-TL overlap based on prediction accuracy. As presented in Figure 2(b), we can notice that when the predictions were accurate, the percentage of high ET-TL overlap cases was relatively higher (57%) compared to when the predictions were incorrect (45%). The percentage of medium ET-TL overlap cases in both scenarios was not considerably different, 21% for incorrect predictions and 18% for

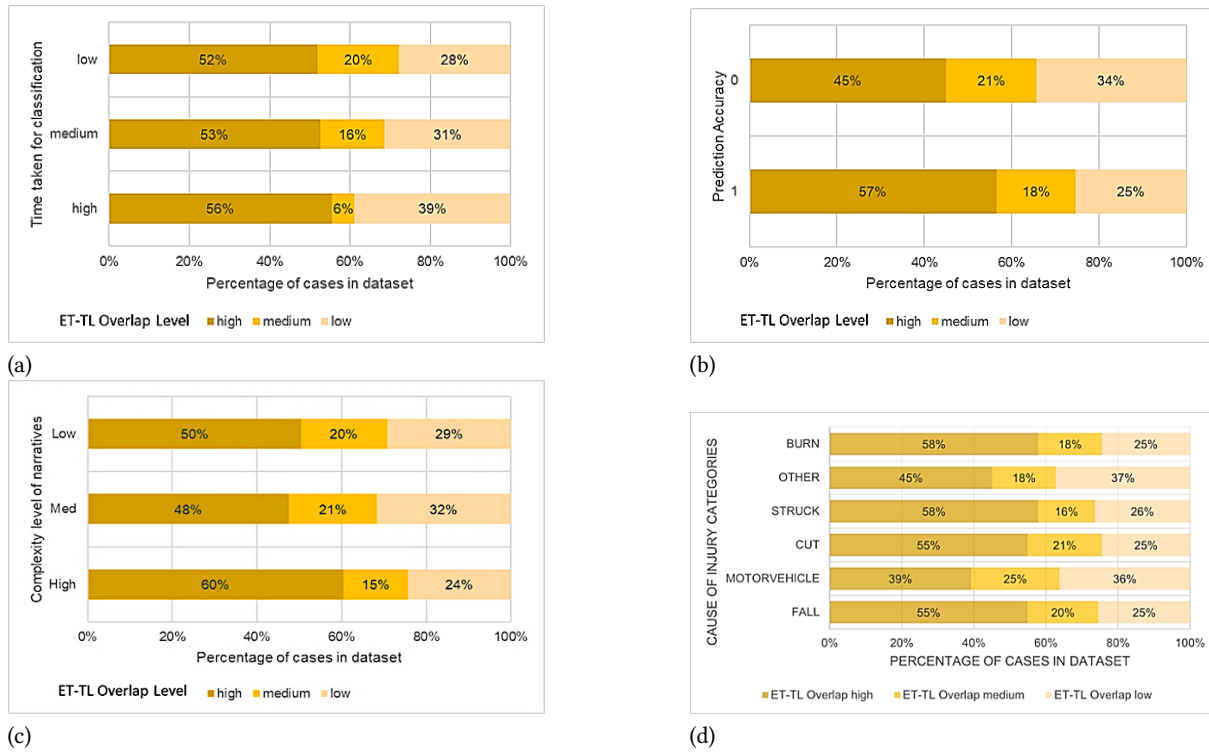


Figure 2: Variation in level of overlap between Eye-tracking and Think-aloud (ET-TL) based on classification accuracy based on (a) Time taken for classification, (b) Prediction accuracy, (c) Complexity level of narratives, and (d) Cause-of-injury categories

correct predictions. The percentage of low ET-TL overlap cases was relatively higher in case of incorrect predictions (34%) as compared to correct predictions (25%). These results suggest that the overlap between eye-tracking and think-aloud was relatively higher when the predictions made by participants were correct. The underlying reason might be that when participants were sure and correct about the cause of injury, the part of the narrative indicating the cause of injury might be clearly mentioned, and the participants would have focused on that part and mentioned the same in their think-aloud recording. Figure 2 (c) shows the distribution of high, medium, and low ET-TL overlap level cases for the different levels of narrative complexity. We can notice that the percentage of high, medium and low ET-TL overlap cases were similar for low and medium complexity groups. It is to be noted that the percentage of high ET-TL overlap cases was considerably higher (60%) for high complexity group as compared to medium (48%) and low (50%) complexity. This might be because for complicated narratives, the participants would have focused more on certain words (resulting in higher FC and FD) which were predictive of the cause-of-injury and thus were also mentioned in the think-aloud data.

Non-parametric statistical tests indicated that most observed differences in ET-TL overlap with respect to narrative complexity, time taken for classification, and prediction accuracy were not statistically significant. Prediction accuracy did not affect ET-TL overlap in SET-1 or SET-2. Complexity was also non-significant in SET-1 and SET-2. In SET-3, however, correct predictions were

associated with significantly greater ET-TL overlap as compared to incorrect predictions (Mann-Whitney $U = 4352.5$, $p = .028$, $r = .15$), and complexity significantly affected ET-TL overlap ($H(2) = 7.765$, $p = .021$). Post-hoc tests with Bonferroni correction ($p < .017$) revealed that High complexity cases had significantly greater ET-TL overlap than Low complexity cases ($U = 2216.5$, $p = .005$), while other pairwise comparisons were non-significant. All remaining contrasts did not reach statistical significance (all $p > .05$).

Next, we compared the ET-TL overlap levels for each of the six cause-of-injury codes, as shown in Figure 2(d). We can note that for categories OTHER and MOTORVEHICLE, relatively higher percentage of low ET-TL overlap level cases was observed as compared to other categories. The other four categories - BURN, STRUCK, CUT, and FALL had similar distribution of high, medium, and low ET-TL overlap cases. The total number of cases for each category was $102 [(3 \text{ data samples collected per prompt set}) * (17 \text{ unique prompt sets}) * (2 \text{ cases of each category in each prompt set})]$. The prediction accuracy for different categories is shown in form of a Confusion Matrix in Figure 3.

As shown in Figure 3, the participants' classification accuracy was best for category BURN, relatively better for FALL and MOTORVEHICLE, followed by CUT, and relatively low for STRUCK and OTHER. It is to be noted that among the different cause-of-injury categories, some had very distinct and clear definitions (e.g., BURN) and hence were easier to classify. On the other hand, some categories had overlapping features which could be confusing for

TRUE CATEGORY	PREDICTED CATEGORY					
	P1: FALL	P2: STRUCK	P3: CUT	P4: BURN	P5: MV	P6: OTHER
C1: FALL	79	14	4	0	1	4
C2: STRUCK	14	49	17	0	6	16
C3: CUT	11	8	71	1	2	7
C4: BURN	2	0	0	96	2	1
C5: MOTORVEHICLE	2	9	2	3	84	2
C6: OTHER	19	6	7	4	3	57

Figure 3: Confusion matrix indicating predictions for different classification categories

the participants to select, e.g., CUT and STRUCK both involve physical contact between humans and objects. The category OTHER was likely to be confusing because it does not have a distinct definition and represents cases where other cause-of-injury codes are inapplicable. These confusions also reflect in the confusion matrix shown in Fig. 3, e.g., frequent confusions between OTHER and STRUCK. Additionally, the relatively low ET-TL overlap level for OTHER (Fig. 2(d)) might be due to unclear narratives causing participants to visually focus on several words but mentioning only few words in their think-aloud reasoning. The reasons behind the relatively low ET-TL overlap observed for MOTORVEHICLE were not clear, as the classification performance did not indicate an increased level of confusion among participants.

4 CONCLUSIONS

This study focused on examining how reliably can eye tracking capture human reasoning during text classification. Overall, we found eye-tracking to be a reliable way of capturing human reasoning during text classification as it yielded reasonable performance (in terms of ET-TL overlap) for varying conditions of narrative complexity, different categories, time taken for classification, and accuracy of prediction. The ET-TL overlap was slightly lower when classification time was higher, which might have been due to those narratives being complicated or confusing. In terms of classification accuracy, the ET-TL overlap was observed to be relatively higher when the predictions made by participants were accurate. In terms of narrative complexity, the ET-TL overlap level was found to be relatively higher for more complex narratives. Some of the limitations of this study were that the sample size was limited to 51 participants and we studied only one type of text classification dataset, which can limit the generalizability of findings from the study. Future work can focus on larger scale user studies with different types of textual data using eye-tracking and possibly other approaches to understand human reasoning during text classification.

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